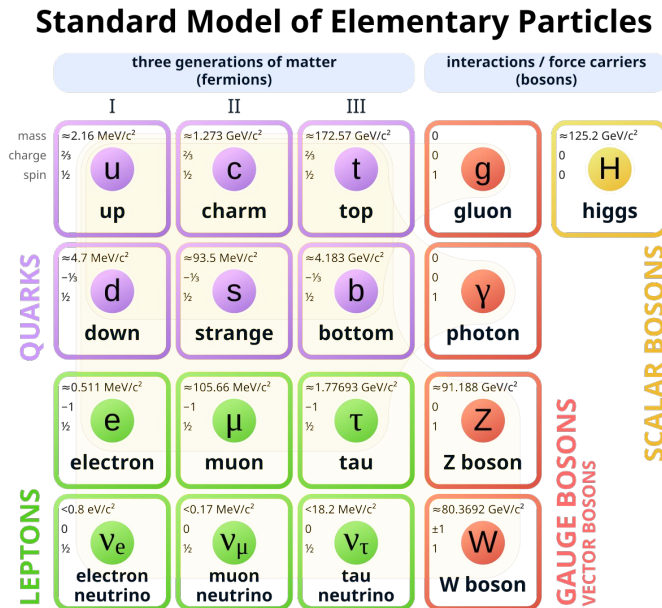


MiniBooNE through the Lens of Machine Learning

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Background

- Neutrinos undergo oscillations
- LSND and MiniBooNE have detected Low Energy Excess (LEE)
- MiniBooNE detected a 4.7σ anomaly (460 unexplained electron-like events)



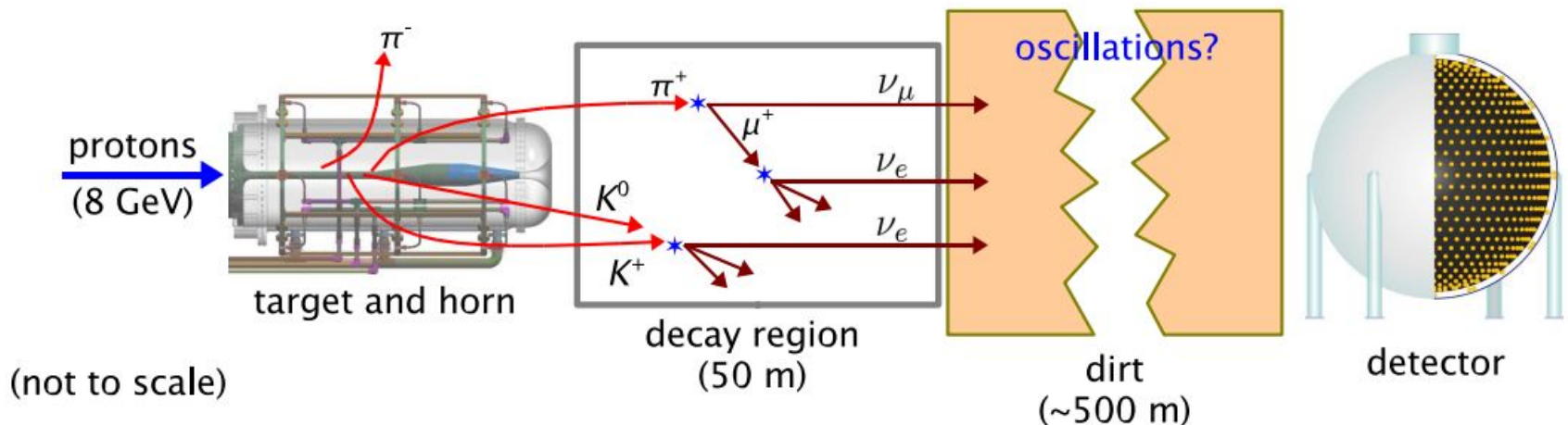
$$\begin{pmatrix} \nu_e \\ \nu_\mu \\ \nu_\tau \end{pmatrix} = \begin{pmatrix} U_{e1} & U_{\mu 1} & U_{\tau 1} \\ U_{e2} & U_{\mu 2} & U_{\tau 2} \\ U_{e3} & U_{\mu 3} & U_{\tau 3} \end{pmatrix} \begin{pmatrix} \nu_1 \\ \nu_2 \\ \nu_3 \end{pmatrix}$$

$$|\nu_e\rangle = U_{e1}|\nu_1\rangle + U_{e2}|\nu_2\rangle + U_{e3}|\nu_3\rangle$$

$$P_{\nu_\alpha \rightarrow \nu_\beta}(t) = |\langle \nu_\beta | \nu_\alpha(t) \rangle|^2$$

Background

- **Protons** hit target and produce **pions**
- $\pi^+ \longrightarrow \mu^+ + \nu_\mu$
- **CCQE:** $\nu_e + n \longrightarrow e^- + p$
- Current analyses rely on 9 sequential selection cuts



Problem:

- The published analysis relies on consecutive selection cuts of reconstructed variables (cutting out a 9-dimensional “signal region” in hyperspace). It thus doesn’t allow for a **cluster-level analysis** of where the excess emerges.
- If the LEE isn’t restricted to the ν_e cluster(s), this indicates an incorrect analysis.
- The reason MiniBooNE has been limited to a selection cuts-based analysis is that much expertise had moved over to MicroBooNE by the time dimensionality reduction (DR) and neural networks (NNs) became commonplace in particle physics.

Motivation:

- We can’t trust, based on Monte Carlo (MC) motivated cuts, that the detected LEE is restricted to one cluster without knowing where the excess lies in hyperspace (or a derivative thereof, e.g. a trustworthy latent space)

Goal (Approaches 1 - 2):

- To find a cleanly separable signal cluster for MC data, and to compare the location and density of this cluster for MC v. experimental data

Overview:

- Use reconstructed quantities such as energy and number of detector hits to analyze the event data.
- Compare the position and density of the “signal cluster(s)” in 2D latent space for MC v. experimental data
- The hope was that we could find a latent space where LEE was a distinctive cluster such that we could analyze how this cluster differs between MC and experimental data

Previous Idea of Subgoals

- 1) Find a latent space for simulated data that places signal in a clean cluster (as little NC π_0 and other backgrounds as possible!)
 - a) *Deciding on DR technique: PCA, LDA, and/or AE*
 - b) *Find appropriate variables to use for DR*
 - c) *Deciding on labeling if we choose supervised approach*
- 2) Quantify the overlap between signal and background (primarily NC π_0) so we can get a sense of the percentage of events that are signal in this cluster
- 3) Use the same DR on experimental data and analyze
 - a) *We should get a cluster centered in ~ the same place as for the simulated data.*
 - b) *Implement appropriate clustering (HDBScan seems appropriate)*
 - c) *Implement statistics to compare counts and location of clusters*
 - d) *Obtain published Particle IDs (PIDs) from the MiniBooNE analysis on both simulated and experimental datasets (see 3.e)*
 - e) *Based on the statistically determined difference in magnitude and shift of the “ ν_e cluster” between simulated and experimental data, determine whether LEE appears to be real*

Intuition:

- i) Suppose there's 2 separate clusters, or a shift of one background cluster
- ii) If PID labels both clusters as signal but they are clearly different clusters, or if published PID cuts up a cluster, this shows PID misaligns with natural clustering
- iii) However, if the latent space for experimental data shows same “miniBooNE cluster location” as that for simulated data, and its magnitude is simply higher, this suggests genuine LEE (or excess of irreducible NC- π_0)

12.1 Selecting ν_e CC QE events

We now turn to the ν_e sample itself. The selection begins with these basic cuts:

- Exactly 1 subevent
- $N_{\text{veto}} < 6$
- $N_{\text{main}} > 200$
- $R_e < 500$ cm
- (endpoint R_μ) < 488 cm

The first eliminates events with apparent muon decays; the second keeps cosmic ray muons out and neutrino events in; the third eliminates Michel electrons; and the last two ensure that the entirety of the event is sufficiently far from the tank wall. (See §10.1.) To complete the ν_e pre-selection, we define an E_ν^{rec} analysis range:

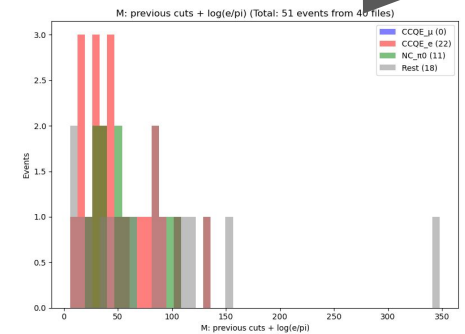
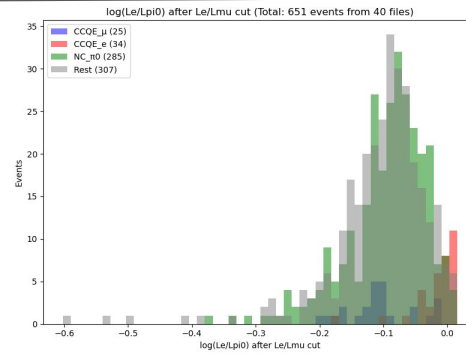
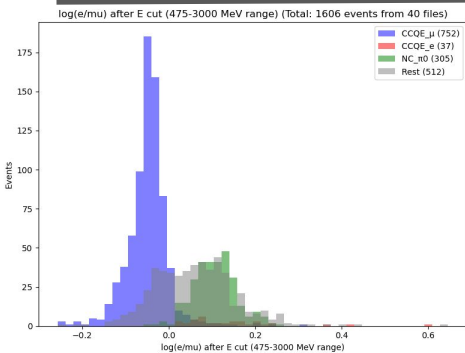
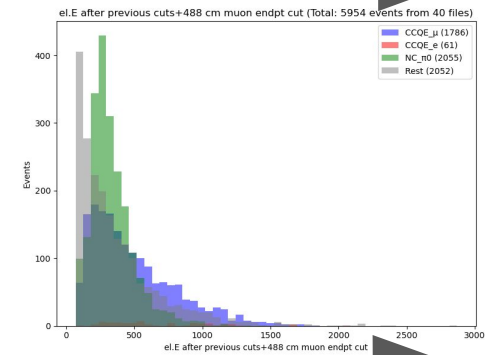
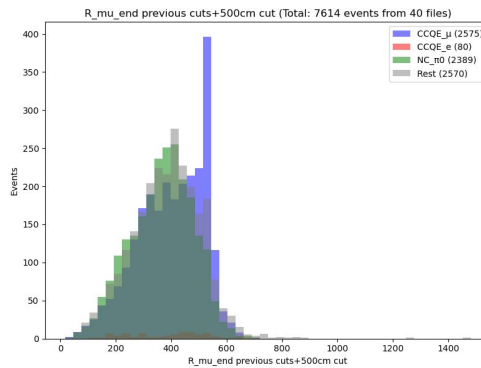
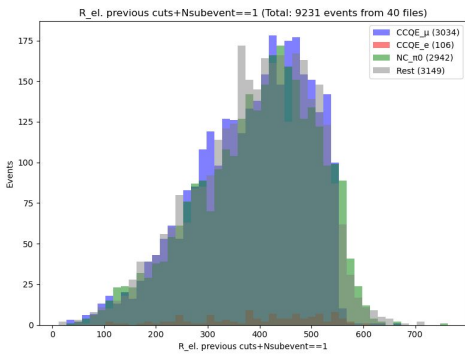
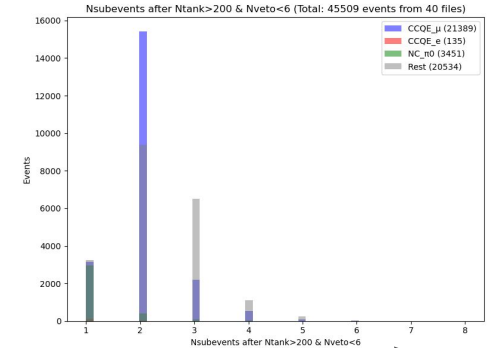
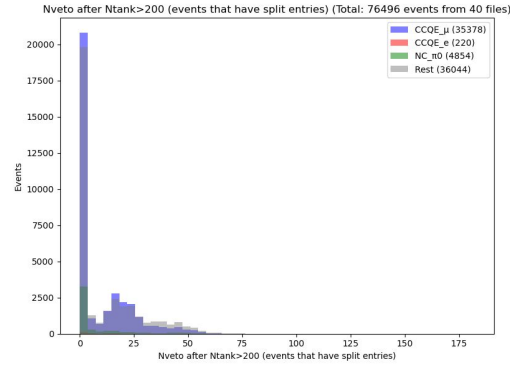
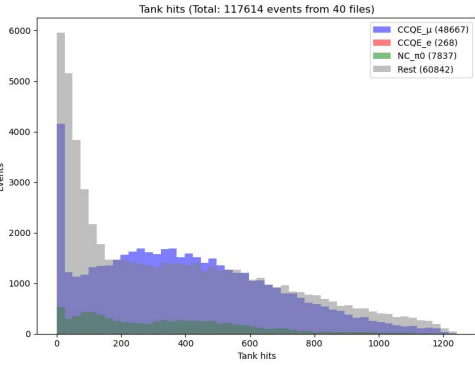
- $475 \text{ MeV} < E_\nu^{\text{rec}} < 3000 \text{ MeV}$.

The lower E_ν^{rec} limit was originally 300 MeV¹, but the threshold was increased to 475 MeV for reasons discussed in §13.2.

The pre-selection is followed by three particle identification (PID) cuts which eliminate π^0 and untagged muon backgrounds. The PID cuts are based on the kinematics and maximum likelihoods returned by the event reconstruction algorithm:

- $\log(\mathcal{L}_e/\mathcal{L}_\mu) > a_0 + a_1 E_e + a_2 E_e^2$
- $\log(\mathcal{L}_e/\mathcal{L}_{\pi^0}) > b_0 + b_1 E_e + b_2 E_e^2$
- $M_{\gamma\gamma} < c_0 + c_1 E_e + c_2 E_e^2$.

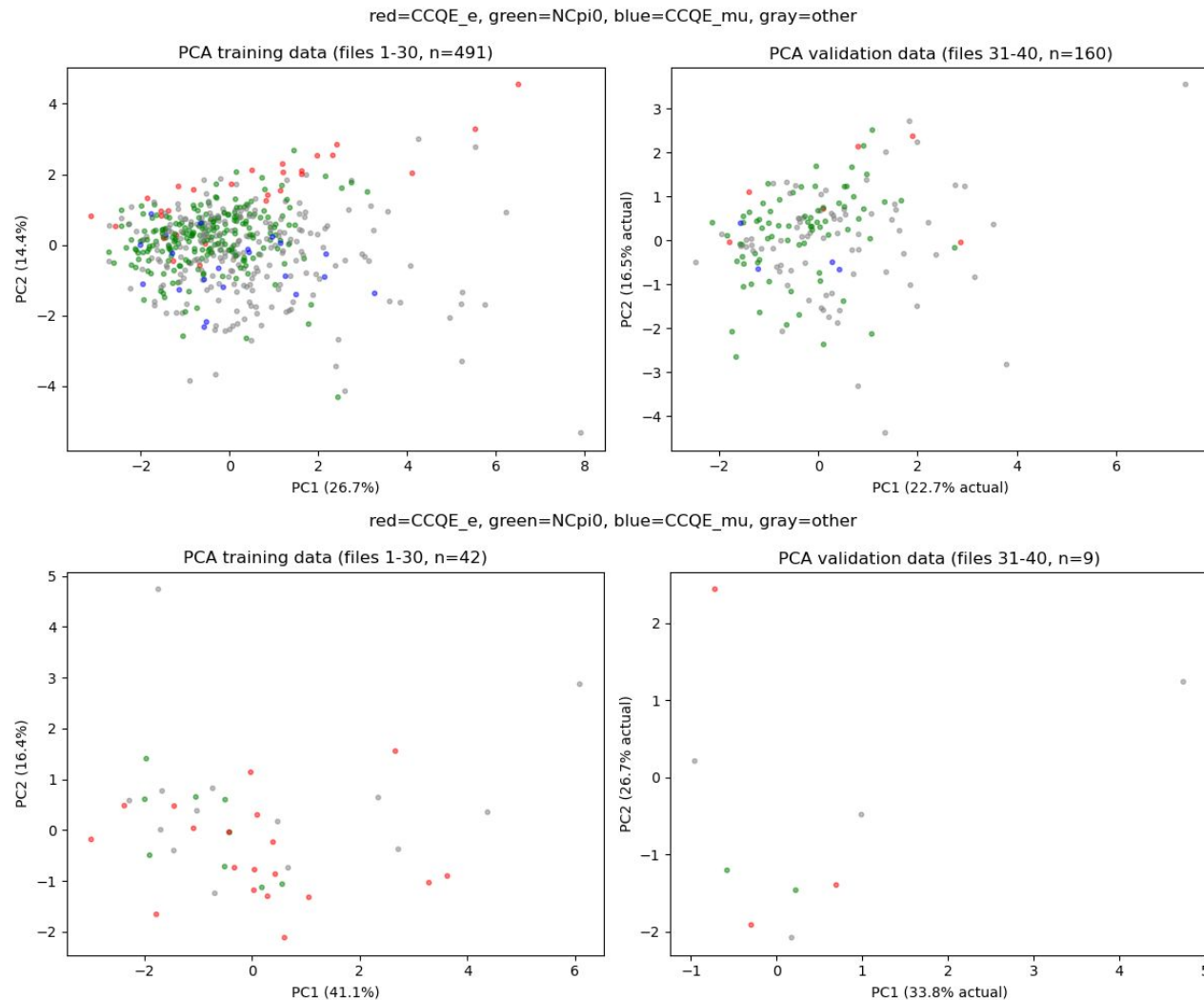
9 Selection Variables after Preceding Cut for MC (Red is Signal)



Third Time is a Charm: Approach 1

- We wanted to see if there was any “natural clustering” of the data using PCA
- There was not (regardless of how many selection cuts we implemented)
- This approach turned out to be naive as we compress non-linear data linearly
- Even when using non-linear methods like AE, there is reconstruction loss

Third Time is a Charm: Approach 1 (typical results)



No separated clusters. TOP: This specific run implements first 7 selection cuts and is based on the 9 selection variables. BOTTOM: Same 9 selection variables but after all 9 selection cuts

Approach 2: Use LDA for Clearer Clustering

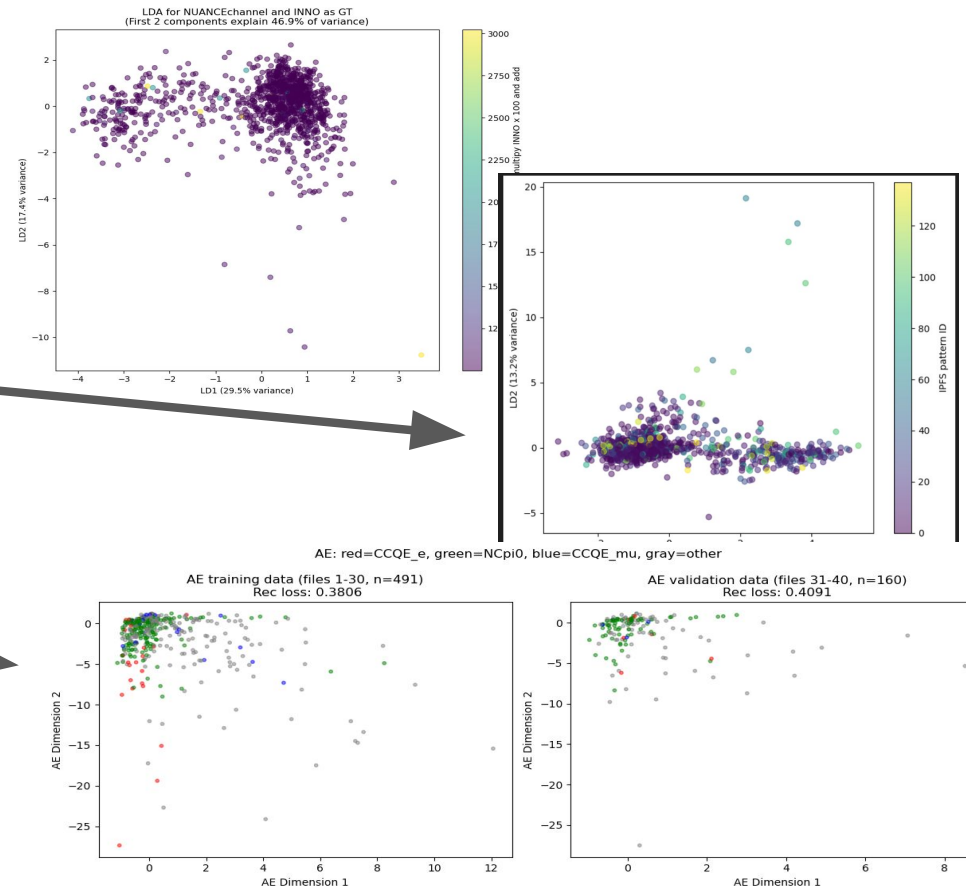
- We've experimented with several labeling schemes:

- “INNO” x “nuance_channel” (Hard to decipher, created hundreds of separate categories)

- IPFS info for the relevant particles (e⁺-, mu⁺-, gamma, pions)

- 4 physics-informed categories: “CCQE_e”, “CCQE_mu”, “NC_pi0”, “other”

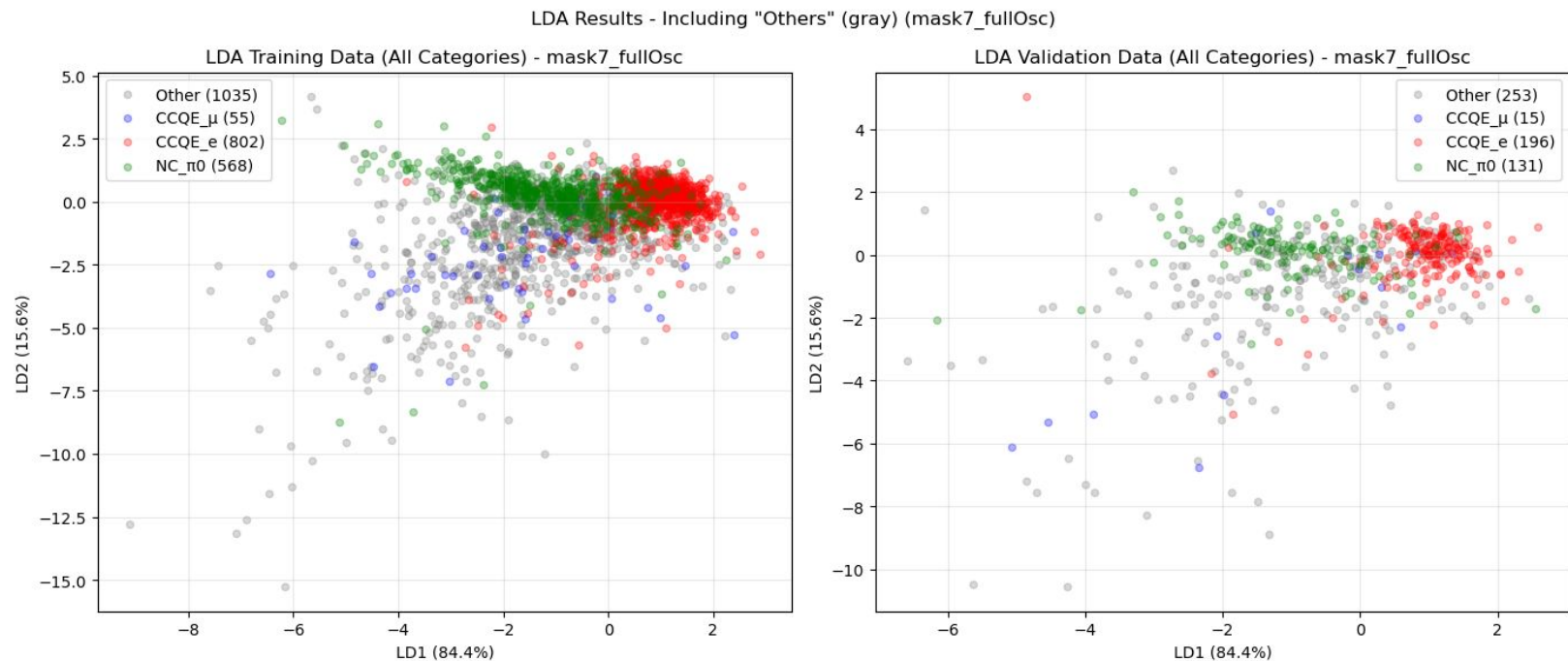
*(INNO = which neutrino type (1-4),
Nuance = decay channel (1-99)
IPFS= Identity-of-particles-final-state)*



Approach 2: Use LDA for Clearer Clustering

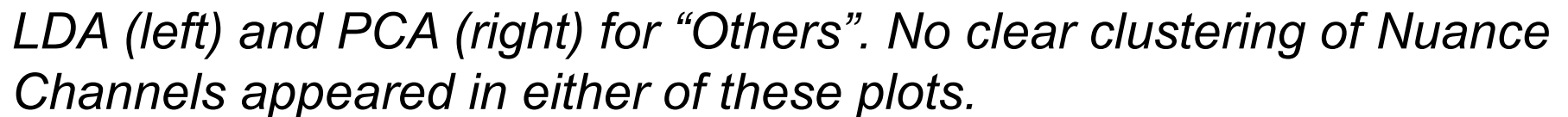
We eventually found a somewhat meaningful latent space

- LDA ran on the non-"Others" categories
- "Others" was "turned back on" *after* the transformation



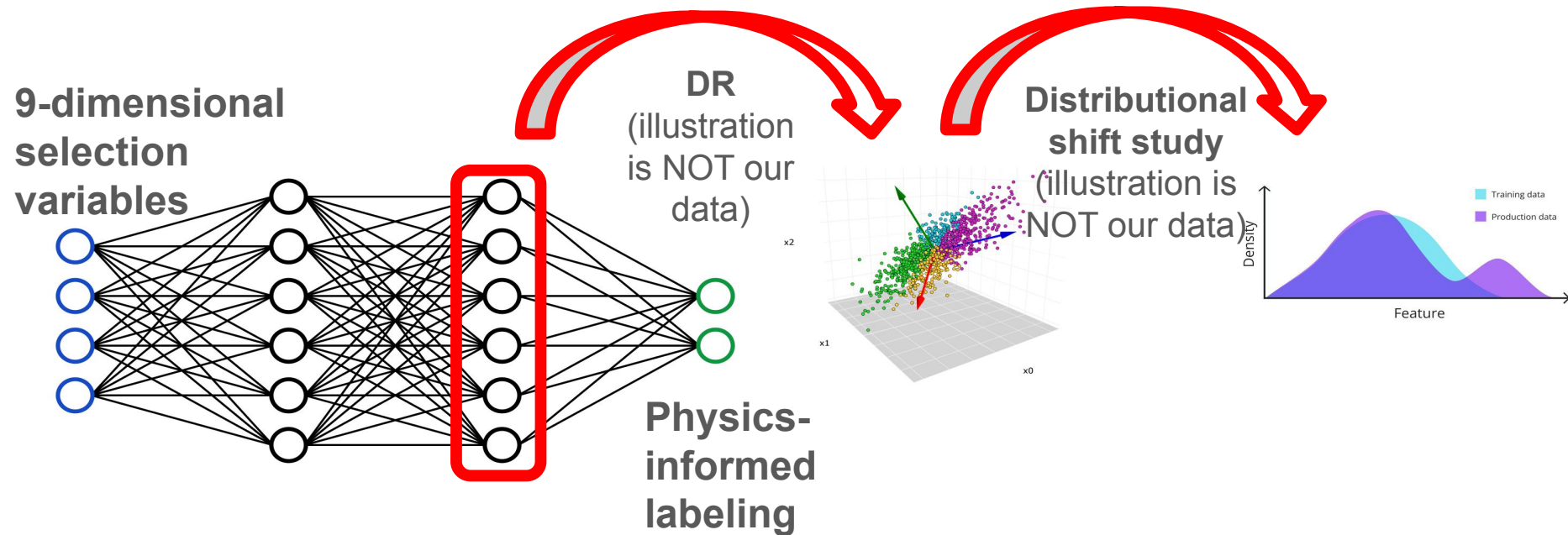
Approach 2: Use LDA for Clearer Clustering

- However, the “Others” category was still heterogeneous, and our labeling scheme was thus naive
- Also, our approach still necessarily entailed information loss (highly non-linear data compressed to 2-dimensions!)



What's Next?

Approach 3: Dimensionality Reduction of Activation Vectors Past the Last Non-Linearity of Classification-Trained NN



Approach 3: Dimensionality Reduction of Activation Vectors Past the Last Non-Linearity of Classification-Trained NN

- **Goal (Approach 3):** To find distributional shifts between MC and experimental data in the principal components of a latent space that is linear to a NN activation space trained on classification of MC events, and to **backtrack the shifts at the event-level to the clusters in this latent space**
- This allows us to observe the distribution of the data in latent space without the problem of information loss
- Unlike selection-cut analyses, this new analysis can observe exactly where the excess lies on the distributions (thus see whether it corresponds to the signal cluster)

Future problems When We Have IPFS Dumps (We Will Test All Options and Keep the Combination that Most Cleanly Separates the Signal)

- We don't know what variables to use for our analysis (although we should initially continue with the 9 selection variables as these are hand-picked to maximize between-class variance)
- We don't know what labels to use (although we recently devised a much more rigorously physics-informed scheme than the ones implemented in Approach 2)* We will benchmark this new scheme v. the 4-classification from Approach 2 and "signal v. background" classification
- Events don't have entries for all variables. Most events (order of 90%) do not have entries for variables associated with later cuts. To deal with this, we must either replace empty entries with e.g. (-1), or still implement cuts 1-7 (which have much higher confidence than cuts 8-9)
- We don't know whether to measure the activation features of the ultimate or penultimate layer of the NN (past the last non-linearity!)
- We don't know whether to use PCA or t-SNE on the activation vectors

- A. A. Aguilar-Arevalo et al. (MiniBooNE Collaboration), Phys. Rev. Lett. 98, 231801 (2007). [Inspire: 748380]
- A. A. Aguilar-Arevalo et al. (MiniBooNE Collaboration), Phys. Rev. Lett. 102, 101802 (2009). [Inspire: 860589]
- A. A. Aguilar-Arevalo et al. (MiniBooNE Collaboration), Phys. Rev. Lett. 105, 181801 (2010). [Inspire: 860589]
- A. A. Aguilar-Arevalo et al. (MiniBooNE Collaboration), Phys. Rev. Lett. 110, 161801 (2013). [Inspire: 1223326]
- A. A. Aguilar-Arevalo et al. (MiniBooNE Collaboration), Phys. Rev. Lett. 121, 221801 (2018). [Inspire: 1675670]

Appendix: New Strategy in High Definition

- **Goal:** Compare MC vs experimental data using distributions for latent space variables that account for non-linear relationships in the data.
- **New approach:**
 - Normalize the data (e.g. -1 to 1) from our 9 dimensions and replace empty values with a (-1)
 - Find out what to do for empty entries, e.g., given a certain criteria, enforce all variables to 0 or 1
 - Implement statistical weighting for the loss since some categories have fewer variables
 - Experiment with MLP architectures (#neurons, #layers, which activation function to use)
 - Find something with low loss on the validation sample
 - Take the second-to-last or last layer (i.e. after the last non-linearity) and project to latent space using PCA or t-SNE. The idea here is that the second-to-last layer contains information that *linearly* maps to the classification. So lowering to 2D here doesn't kill information the way that our previous PCA/LDA did
 - Create binned histograms out the data based on where it exists in the 2 (or 3) latent dimensions
 - Calculate distributional shifts for simulated vs experimental data - hopefully distributions do not look the same, indicating some fundamental problem with the simulation
- **Notes:**
 - We should overengineer; too small MLP won't fit all the structure of the data
 - This code is not hard to write. The project is rather: Experiment with different architectures, which layer to look at, find some combo with low loss
 - **This approach has never been before been implemented on MiniBooNE. It could be that we discover distributional shifts in the principal components that show systematic MC mismodelings. This is exciting.**

Appendix: New Labeling Scheme

The next step will be to implement the following labeling scheme:

- 1: ν_e CC single $e^\pm$ with no other particles above $\check{C}$ threshold
- 2: All other ν_e CC ($e^\pm$ + any other particles)
- 10: Events with 1γ (e.g. radiative Δ), no other particles above $\check{C}$ threshold
- 11-13: Single π^0 events where only one γ is visible, no other particles above $\check{C}$ threshold
- 21, 22: Single π^0 events where both γ are in principle visible, no other particles above $\check{C}$ threshold
- 3: Single $\mu^\pm$ events with no other particles above $\check{C}$ threshold (including decay e)
- 4: Single $\pi^\pm$ events with no other particles above $\check{C}$ threshold (including decay e)
- 5: Events with no particles above $\check{C}$ threshold
- 6: (Everything else)

ID			Criteria (IPFS, PFSP)	
1	A	Primary $e^\pm$	IPFS[0] = $e^\pm$	
	B	No visible π	No π^0 or photons from π^0 No $\pi^\pm$ with $T > 100$ MeV No decay $e^\pm$	
	C	No visible protons	No p with $T > 350$ MeV	
	D	OR No Č light	NHTTCER < 10 for all particles except primary $e^\pm$	

ν_e CC events with no detectable final state particle apart from primary $e^\pm$

- A & (B&C) or A & D

All other ν_e CC

ID			Criteria (IPFS, PFSP)	
2	A	Primary $e^\pm$	IPFS[0] = $e^\pm$	
	B		Complement of $0\pi 0p$	
	C	OR $\checkmark$ light	NHTTCER > 10 for any particle besides primary $e^\pm$	

- A & B or A & C

Single γ

ID			Criteria (IPFS, PFSP)	
10	A	Single γ	Exactly one γ with $E_\gamma > 100$ MeV	
	B	No visible primary lepton	IPFS[0] is a neutrino OR IPFS[0] is not an $e^\pm$ OR IPFS[0] is $\mu^\pm$ with $T < 100$ MeV, no decay $e^\pm$	
	C	No visible π	No π^0 or photons from π^0 AND No $\pi^\pm$ with $T > 100$ MeV AND No decay $e^\pm$	
	D	No visible protons	No p with $T > 350$ MeV	
	E	OR No Č light	NHTTCER < 10 for all particles except primary γ	

Events with intrinsically only one γ produced with no other visible particles

- A & (B&C&D) or A&E

Single γ π^0 events

ID			Criteria (IPFS, PFSP)
	A0	Single π^0	Exactly one π^0
11 12 13	A1 A2 A3	1 visible γ	1 decay photon with $E < 20$ MeV OR 1 decay photon with $VRTX > 450$ cm OR 1 decay photon with $NHTTCER < 10$
	B	No visible primary lepton	IPFS[0] is a neutrino OR IPFS[0] is not an $e^\pm$ OR IPFS[0] is $\mu^\pm$ with $T < 100$ MeV, no decay $e^\pm$
	C	No additional visible π	No additional π^0 or photons from π^0 AND No $\pi^\pm$ with $T > 100$ MeV AND No decay $e^\pm$
	D	No visible protons	No p with $T > 350$ MeV
	F	OR No Č light	$NHTTCER < 10$ for all particles except primary decay γ

- 11
 - A0 & A1 & (B&C&D) or
 - A0 & A1 & F,
- 12
 - A0 & A2 & (B&C&D) or
 - A0 & A2 & F
- 13
 - A0& A3 & (B&C&D) or
 - A0 & A3 & F

Events where π^0 produced two γ s but one is undetectable

Single π^0 events

ID			Criteria (IPFS, PFSP)
	A0	Single π^0	Exactly one π^0 with $E_\gamma > 20$ MeV for both photons
21 22	A1 A2	1 π^0 from initial state 1 π^0 from secondary	π^0 appears in the IPFS stack π^0 identified through decay gammas
	B	No visible primary lepton	IPFS[0] is a neutrino OR IPFS[0] is not an $e^\pm$ OR IPFS[0] is $\mu^\pm$ with $T < 100$ MeV, no decay $e^\pm$
	C	No additional visible π	No additional π^0 or photons from π^0 AND No $\pi^\pm$ with $T > 100$ MeV AND No decay $e^\pm$
	D	No visible protons	No p with $T > 350$ MeV
	E	OR No Č light	NHTTCER < 10 for all particles except primary decay γ s

- 31
 - A0 & A1 & (B&C&D) or
 - A0 & A1 & E
- 32
 - A0 & A2 & (B&C&D) or
 - A0 & A2 & F

Events where π^0 produced two γ s and both are in principle detectable

Single μ events

ID			Criteria (IPFS, PFSP)
3	A	Single μ	IPFS[0] = $\mu^\pm$ with $T > 100$ MeV
	C	No additional visible π	No π^0 or photons from π^0 AND No $\pi^\pm$ with $T > 100$ MeV AND No decay $e^\pm$
	D	No visible protons	No p with $T > 350$ MeV
	E	OR No Č light	NHTTCER < 10 for all particles except primary $\mu^\pm$

- 4
 - A & (C&D) OR
 - A & (E)

Events where a single $\mu^\pm$ is produced

Single $\pi^\pm$ events

ID			Criteria (IPFS, PFSP)
4	A	Single $\pi^\pm$	IPFS[0] = $\pi^\pm$ with $T > 100$ MeV
	B	No visible primary lepton	IPFS[0] is a neutrino OR IPFS[0] is not an $e^\pm$ OR IPFS[0] is $\mu^\pm$ with $T < 100$ MeV, no decay $e^\pm$
	C	No additional visible π	No π^0 or photons from π^0 AND No other $\pi^\pm$ with $T > 100$ MeV AND No decay $e^\pm$
	D	No visible protons	No p with $T > 350$ MeV
	E	OR No Č light	NHTTCER < 10 for all particles except primary $\pi^\pm$

- 5
 - A & (B&C&D) OR
 - A & (E)

Events where a single $\pi^\pm$ is produced

No Cherenkov

ID			Criteria (IPFS, PFSP)
5			
	B	No visible primary lepton	IPFS[0] is a neutrino OR IPFS[0] is not an $e^\pm$ OR IPFS[0] is $\mu^\pm$ with $T < 100$ MeV, no decay $e^\pm$
	C	No visible π	No π^0 or photons from π^0 AND No other $\pi^\pm$ with $T > 100$ MeV AND No decay $e^\pm$
	D	No visible protons	No p with $T > 350$ MeV
	E	OR No Č light	NHTTCER < 10 for all particles

Events where no significant Cherenkov radiation is produced by any particle

- (B & C & D) or E